CA Assignment01 REPORT

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2022330

Part01: Simulation Time

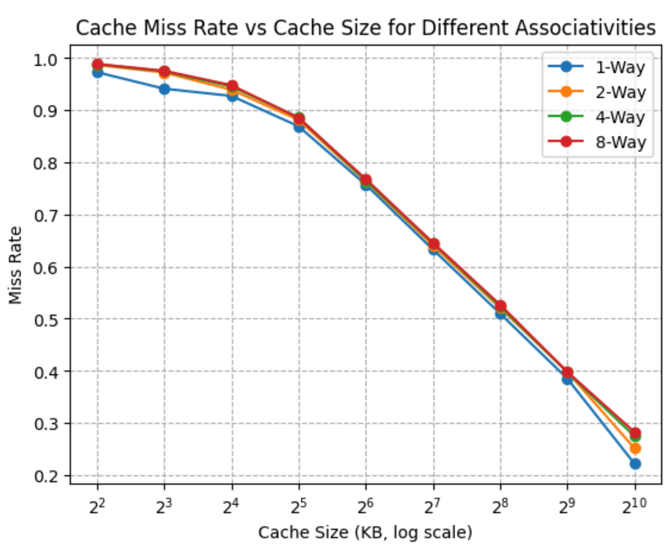
RiscvTimingSimpleCPU() : 54869602000

RiscvO3CPU(): 9339759000

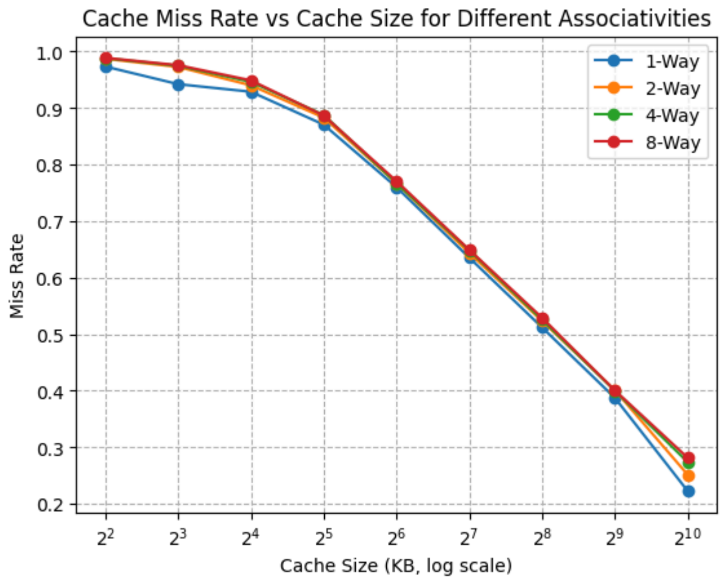
Part02: Cache size vs Cache Miss Rate

- **L2 Cache sweep keeping L3 cache same:**

  RiscvTimingSimpleCPU:

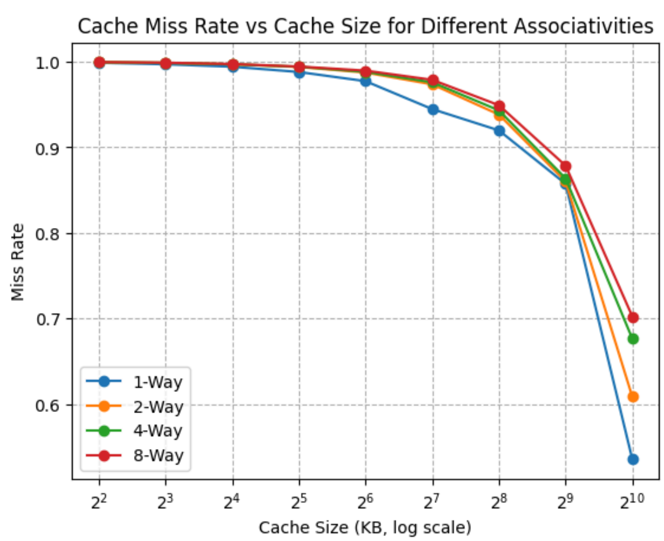


RiscvO3CPU():

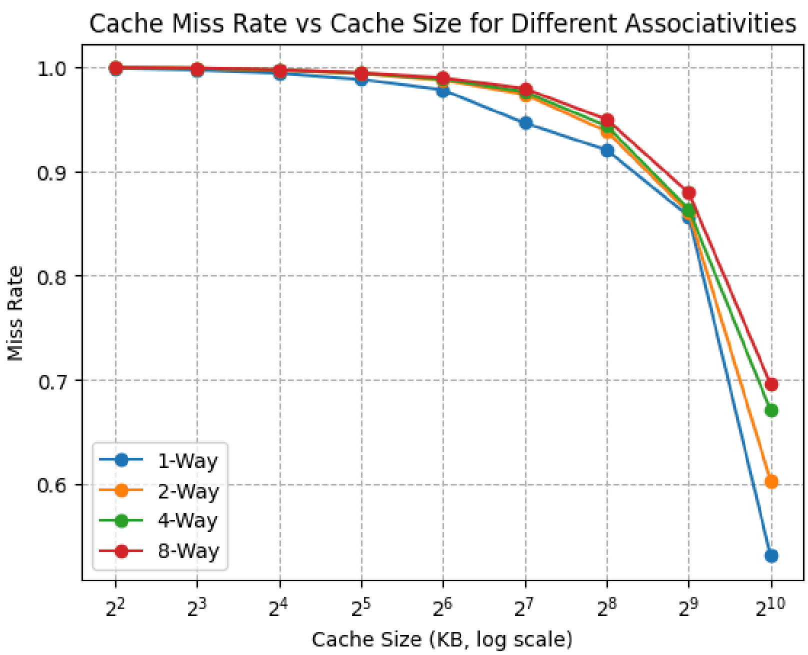


- **L3 Cache sweep keeping L2 same:**

RiscvTimingSimpleCPU:



RiscvO3CPU:



Conclusion:

- **Part01: O3CPU simulation takes significantly fewer ticks, showing its efficiency in handling instruction execution compared to the TimingSimpleCPU.**

- **L2 Cache: The observations indicate a strong relationship between L2 cache size, associativity, and the resulting miss rate. Increasing the L2 cache capacity consistently lowers the miss rate, leading to better performance. However, choosing an excessively large cache may not yield benefits**

proportional to its cost. Although larger cache sizes and higher associativity do contribute to reducing misses, the gains gradually diminish beyond a certain point and are also influenced by the program's memory access behavior.

- **For the L3 cache, enlarging its size and increasing associativity also reduces the miss rate, but the reduction is more gradual compared to L2. This is because L3 only processes the accesses that have already missed in both L1 and L2. Such accesses generally exhibit weaker locality, making them more difficult to retain in the cache. Consequently, even a large L3 cache does not deliver the sharp performance gains observed with L2.**